

Name:

Collaborators:

**Instructions:** Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

## Choose an Integrating Factor

### 1. The Reverse Product Rule

The following technique applies the *reverse product rule* to find the *general solution* to a 1st-order linear ODE.<sup>1</sup> The tasks below solves the ODE  $\frac{dy}{dt} - y = -t$ .

| Task                                                                                                                                                                                                                      | Show work here  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Use the product rule to expand $(yu)'$ .<br>_____                                                                                                                                                                         | Line 0<br>_____ |
| In the given ODE, write $\frac{dy}{dt}$ as $y'$ .<br>_____                                                                                                                                                                | Line 1<br>_____ |
| Notice that the left-hand side of the equation in Line 1 looks a lot like the expanded product rule but is missing the function $u$ . So multiply both sides by $u$ , a function that we will determine shortly.<br>_____ | Line 2<br>_____ |
| Because, so far, $u$ is an arbitrary function, we can have $u$ satisfy any differential equation that we want. Use $u' = -u$ to rewrite the left-hand side of Line 2 to look like Line 0.<br>_____                        | Line 3<br>_____ |
| Solve $u' = -u$ .<br>_____                                                                                                                                                                                                | Line 4<br>_____ |
| Replace $u$ in the equation from Line 2 with your solution from Line 4.<br>_____                                                                                                                                          | Line 5<br>_____ |
| Show that the equation in Line 5 can be rewritten as $(ye^{-t})' = -te^{-t}$<br>_____                                                                                                                                     | Line 6<br>_____ |

<sup>1</sup>This problem continues in the next page.

| Task                                                                                                                          | Show work here |
|-------------------------------------------------------------------------------------------------------------------------------|----------------|
| Write integrals with respect to $t$ on both sides of the equation in Line 6. Then, apply the Fundamental Theorem of Calculus. | Line 7         |
| Obtain the <i>general solution</i> .                                                                                          | Line 8         |

The function  $u$  is called the *integrating factor*, a function chosen so that when the entire equation is multiplied by it, the left-hand side becomes the derivative of a product. This allows the equation to be integrated directly using the *reverse of the product rule*.

## 2. The Process of Integrating Factors

The method of *Integration Factors (IF)* is used to find the *general solution* of an ODE written in *1st-order linear form*,

$$\frac{dy}{dt} + py = q,$$

where  $p$  and  $q$  are continuous functions of  $t$ .

Follow the tasks below to see how the method works to solve for the exact solution of  $\frac{dy}{dt} = 1 - \frac{y}{1-t}$ .

| Task                                                                                                                                 | Show work here |
|--------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Rewrite the given ODE into <i>1st-order linear form</i> . Then identify $p$ and $q$ .                                                |                |
| Determine the <i>integrating factor</i> $u = e^{\int p dt}$ . Note that this is the solution to the subproblem $u' = up$ .           |                |
| Multiply $u$ in both sides of the ODE, then apply the <i>reverse product rule</i> :<br>$uy' + upy = uq$ $(uy)' = uq.$                |                |
| Integrate both sides with respect to $t$ of the equation using the Fundamental Theorem of Calculus:<br>$\int (uy)' dt = \int uq dt.$ |                |
| Solve for the <i>general solution</i> .                                                                                              |                |